
AI-102

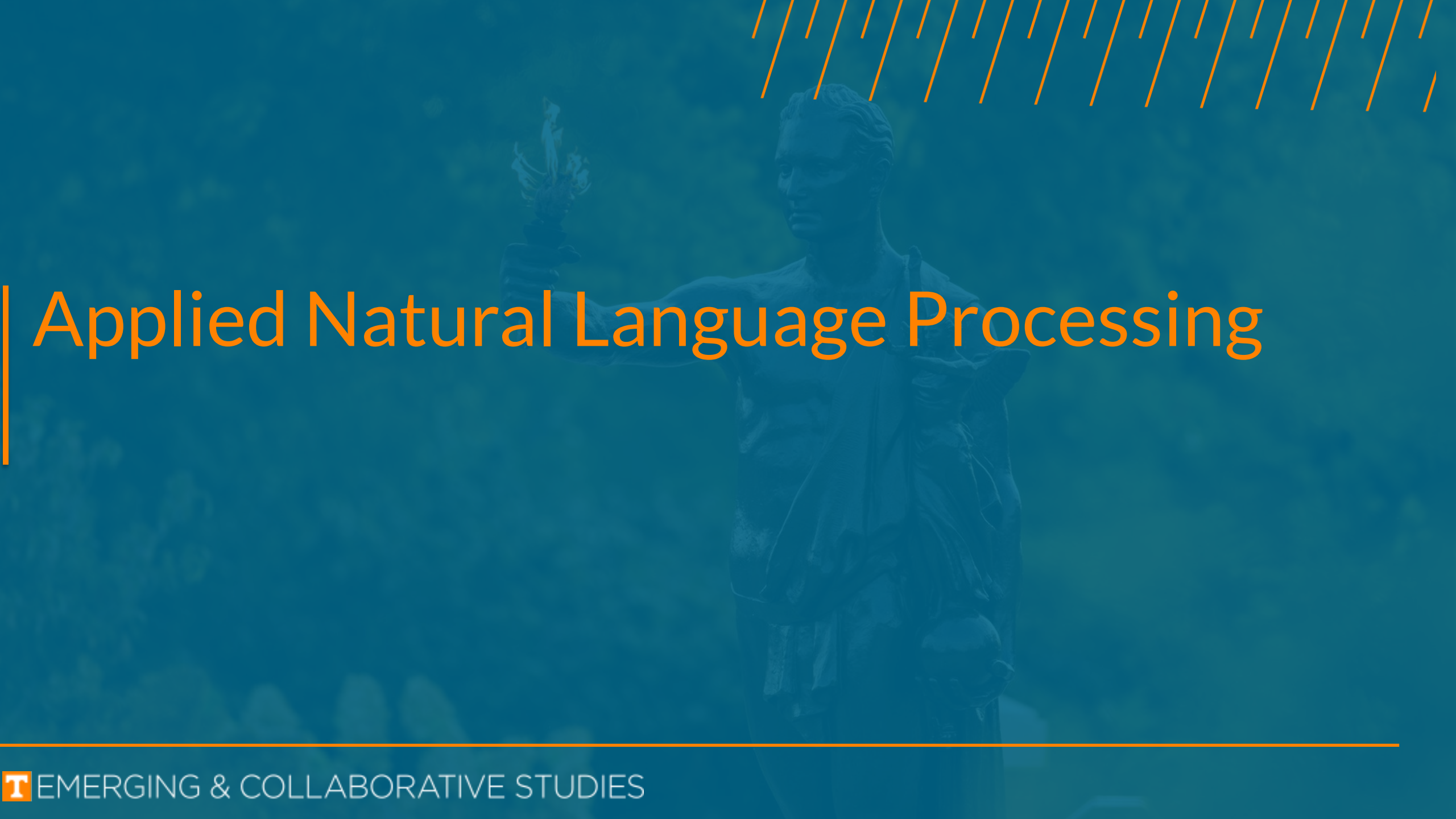
Natural Language Processing (NLP)

Dr. Jimmy Ardiansyah



EMERGING &
COLLABORATIVE STUDIES





Applied Natural Language Processing

Practical NLP in Social Media

- Platforms: Twitter, Facebook, Instagram, WhatsApp, etc.
- Huge volume, velocity, variety of posts every second
- Mostly unstructured text → impossible to read manually
- NLP is needed to:
 - **Trending topic detection** – find hashtags/topics getting the most traction
 - **Opinion mining** – summarize public opinions about products, services, policies
 - **Sentiment detection** – track positive/negative mood over time for brands
 - **Rumor / fake news detection** – flag false or misleading content
 - **Adult / toxic content filtering** – remove profanity, threats, racism, etc.
 - **Customer support** – prioritize and respond to complaints via social channels



Practical NLP in E-Commerce

- Customers talk in natural language: questions, complaints, reviews
- Huge volume of text from chat, email, and social media
- NLP helps you understand intent, mood, and product needs
- Results: faster support, higher sales, better customer satisfaction
- NLP is needed to:
 - **Customer support chatbots:** answer questions in real time
 - **Product search and FAQ:** understand natural language queries
 - **Recommendations:** match user language to product catalog
 - **Review and rating analysis:** detect sentiment and common issues
 - **Engagement:** status, returns, shipping questions



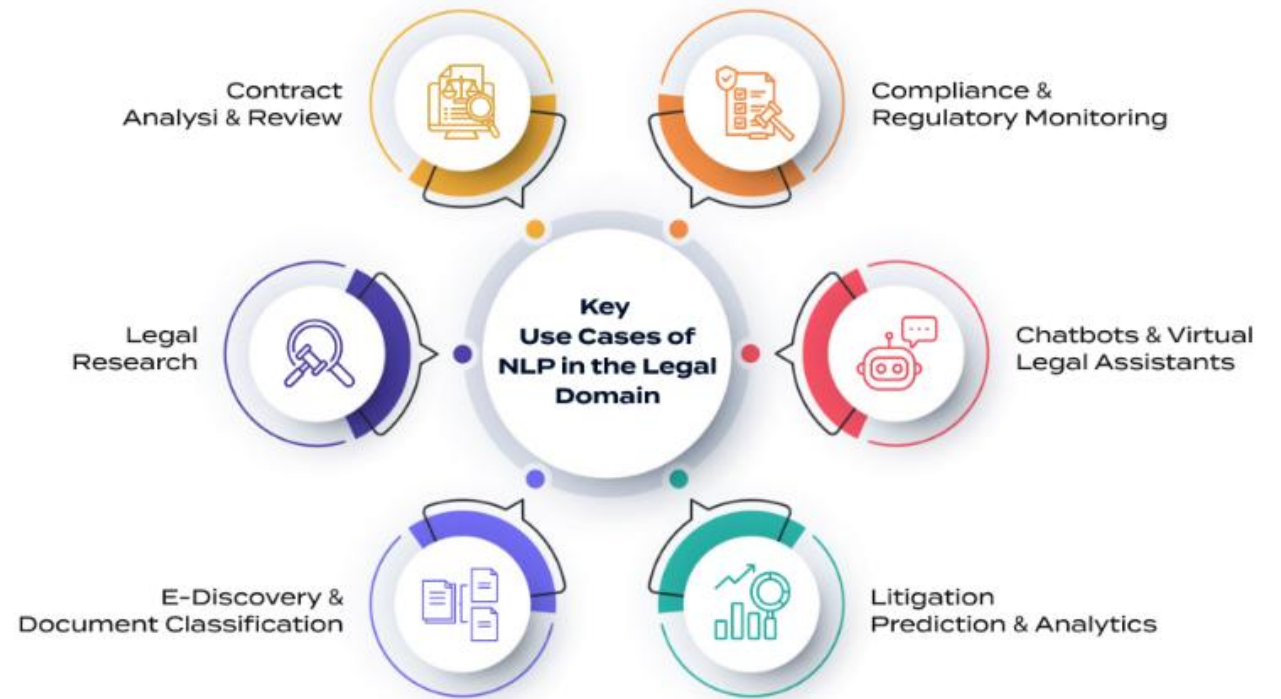
Practical NLP in Healthcare

- Healthcare generates huge amounts of text every day:
- Clinical notes, discharge summaries, radiology reports
- Patient messages, call-center logs, telemedicine chats
- Most of it is unstructured free text → hard to search or analyze
- NLP helps turn text into structured, actionable information
- NLP is needed to:
 - Medical imaging analysis
 - Patient engagement
 - Drug discovery and development
 - Personalize medicine
 - Remote monitoring and telemedicine
 - Clinical documentation analysis
 - Fraud detection and claims processing



Practical NLP in Law/Legal

- Law is text heavy
- Statutes, regulations, case law, contracts, brief emails
- Thousands of pages per case, impossible to read everything manually
- Deadlines and high stakes require quick and accurate insight
- NLP helps convert legal text into searchable, structured, and summarized information
- NLP is needed to:
 - Contract analysis and review
 - Compliance regulatory monitoring
 - Chatbots and virtual legal assistants
 - Litigation prediction and analytics
 - E-discovery document classification
 - Legal research



Practical NLP in Finance

- Finance runs on information, especially text
- News, earnings reports, analyst notes, social media
- Huge volume of documents every day, hard to read manually
- Decisions often need to be fast and data driven
- NLP turns unstructured text into signals and structured data
- NLP is needed to:
 - Onboarding and document processing
 - Fraud detection
 - Regulatory compliance
 - Credit scoring
 - Customer retention
 - Payment
 - Investing
 - Anomaly detection



**Onboarding and
Document Processing**



**Fraud
Detection**



**Regulatory
Compliance**



**Credit
Scoring**



**Customer
Retention**



Payments



Investing



**Anomaly
Detection**

Practical NLP in Energy

Energy sector uses computer analysis of human language to extract insights from unstructured text and voice data for applications like predictive maintenance, demand forecasting, and customer service. By processing documents, reports, maintenance logs, and customer feedback, NLP helps the industry become more efficient, optimize operations, improve customer engagement, and support sustainability goals.

- NLP is needed to:
 - Predictive maintenance
 - Data extraction
 - Customer service
 - Market analysis
 - Operational efficiency
 - Research and development
 - Compliance monitoring



Practical NLP in Education

Education sector uses natural language processing (NLP) to enhance learning, with applications including automated essay scoring, providing real-time feedback on writing, personalizing e-learning experiences, and assisting with language acquisition. It can improve student engagement, support struggling readers, help teachers automate tasks, and offer personalized instruction by analyzing and processing human language.

- NLP is needed to:
 - Automated writing feedback
 - Personalized learning
 - Language learning assistance
 - Reading support:
 - Assessment and evaluation
 - Intelligent Tutoring Systems (ITS)



Practical NLP in Transportation

Natural Language Processing (NLP) is used to analyze and interpret human language data such as written text, spoken voice, and digital messages to help transportation systems operate more efficiently. By understanding how people communicate, NLP helps improve road and passenger safety, enhance customer service experiences, optimize daily operations, and support smarter decision-making across the entire transportation sector.

- NLP is needed to:
 - Traffic & Incident Detection
 - Voice Control in Vehicles
 - Safety & Accident Analysis
 - Fleet Maintenance & Logistics
 - Smart City Transit Planning
 - Customer Service Automation







EMERGING &
COLLABORATIVE
STUDIES
